Investor variance report

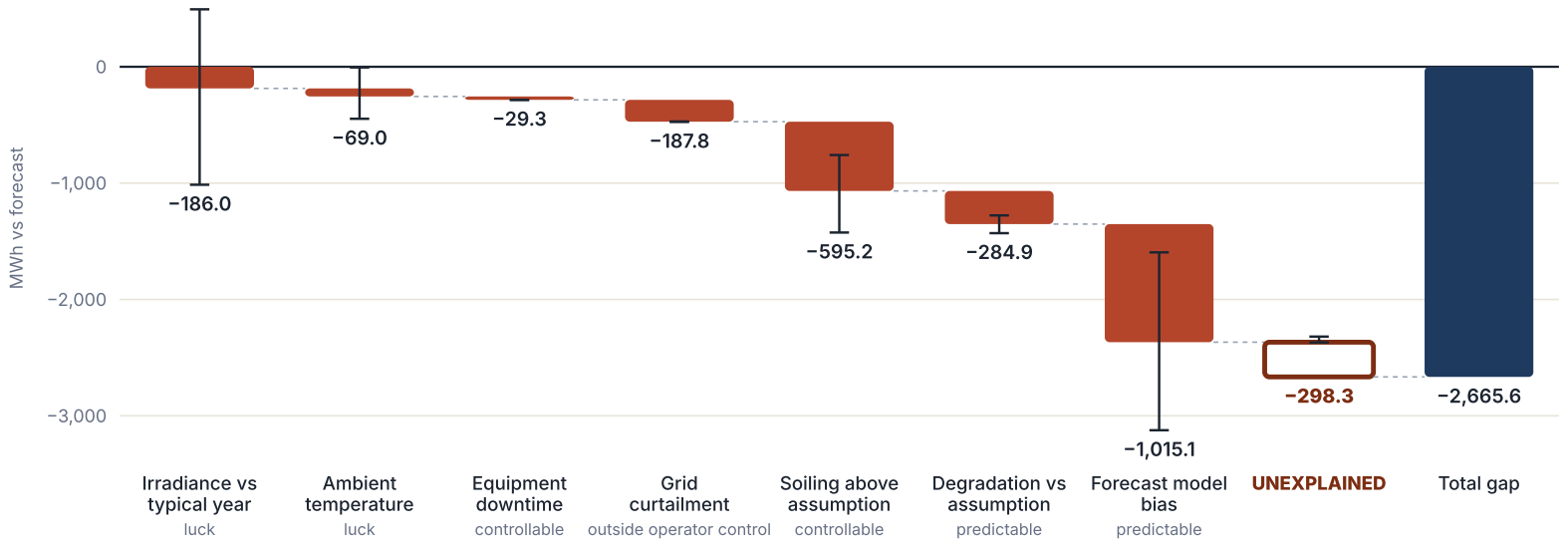
Synthetic 30 MWp plant, Jodhpur region, Rajasthan · FY 2025-26 (Apr 2025 - Mar 2026) · model vareng-0.1.0 · **SYNTHETIC DATA**

Summary

In FY 2025-26 (Apr 2025 - Mar 2026), the synthetic plant delivered 48,210 MWh against a forecast of 50,875 MWh: 2,666 MWh (5.2%) below forecast, worth about INR 72.0 lakh at the assumed tariff. Forecast model bias cost 1,015.1 MWh [predictable, low confidence]. Soiling above assumption cost 595.2 MWh [controllable, low confidence]. Degradation vs assumption cost 284.9 MWh [predictable, medium confidence]. Grid curtailment cost 187.8 MWh [outside operator control, high confidence] Potentially claimable; needs contract review. 298.3 MWh remains unexplained and is larger than measurement uncertainty can account for (90% band -300.7 to -250.3 MWh).

Every number in this paragraph is machine-checked against engine output: passed ✓.

Waterfall



Causes

Cause	Class	MWh	₹ lakh	90% band (MWh)	Confidence
Irradiance vs typical year	Luck	−186.0	−5.0	−1,014 to 494	low
Ambient temperature	Luck	−69.0	−1.9	−261 to 180	low
Equipment downtime	Controllable	−29.3	−0.8	−29 to −29	high
Grid curtailment (potentially claimable)	Outside operator control	−187.8	−5.1	−189 to −186	high
Soiling above assumption	Controllable	−595.2	−16.1	−953 to −287	low
Degradation vs assumption	Predictable	−284.9	−7.7	−363 to −210	medium
Forecast model bias	Predictable	−1,015.1	−27.4	−1,772 to −243	low
Unexplained	–	**−298.3**	−8.1	−301 to −250	low
Total gap		**−2,665.6**	−72.0		

Unexplained

298,333 kWh of the gap is NOT explained by the data we hold. It is larger than input uncertainty alone can produce (90% band -300,725 to -250,262 kWh).

Data that would reduce it: String-level current logs / combiner-box data (finds partial-string failures); Thermal imaging or an IV-curve scan of suspect arrays; A shading survey of new structures or vegetation near the site; Inverter derating and MPPT-fault event codes not currently in the status log.

Method and caveats

Order-independent (Shapley) attribution over a 7-switch plant simulator; unexplained = metered − simulated; 90% bands from 80 Monte Carlo runs. Forecast-model derate calibrated on 93 hours (2025-04-01 to 2025-04-21). SYNTHETIC DATA: plant, outages, curtailment and weather are simulated (weather shaped like Rajasthan, not measured). Irradiance sensor: 40 hours missing and gap-filled with forecast-year values. Soiling comes from a soiling station with unknown calibration bias. Tariff INR 2.70/kWh is an assumption, not a PPA figure. Curtailment profile is illustrative, shaped like reported 2026 patterns. Curtailment flags are not legal conclusions.